Advanced Causal Inference

Lecture 04: Observational Study

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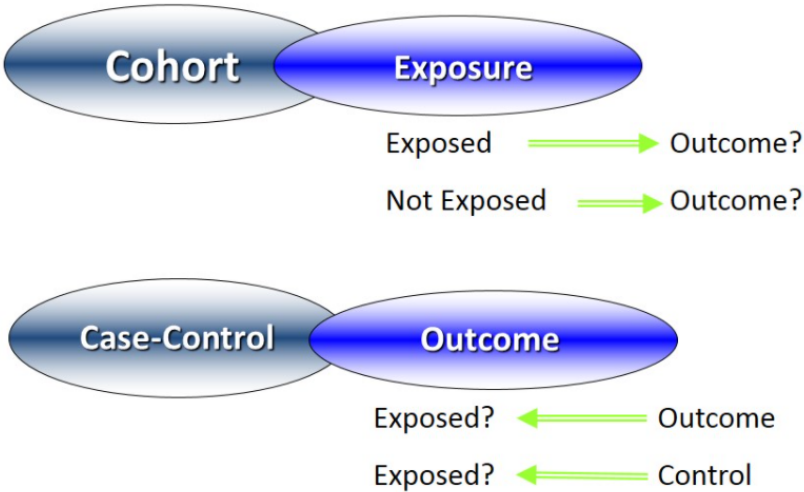
Outline

1. Observational Study
2. Confounding
3. Causal Assumptions
4. Identification
5. Propensity Score & Balancing Score

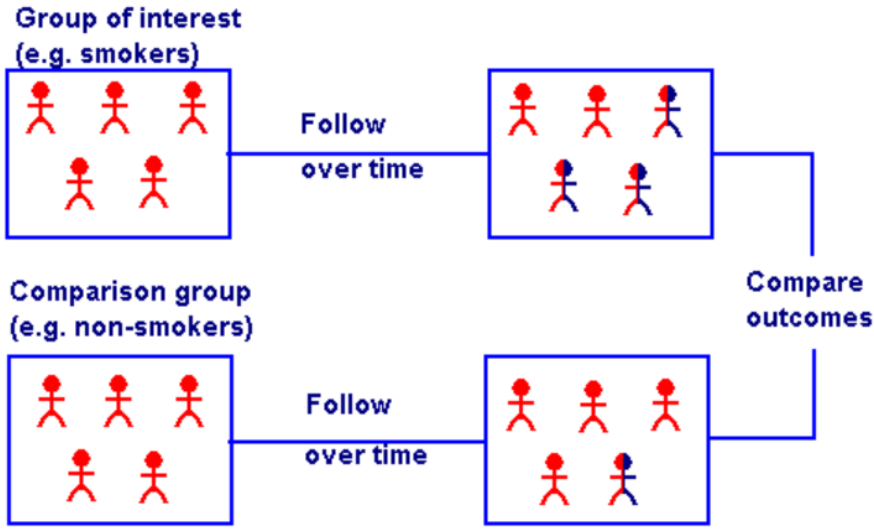
Observational study

- Observational study: researchers simply observe or question the subjects
- Randomized experiments: researchers randomly assign the values of an action A to the subjects, and observe the effects of the action on a response variable Y

Types of Observational Studies



Cohort Study



Cohort study

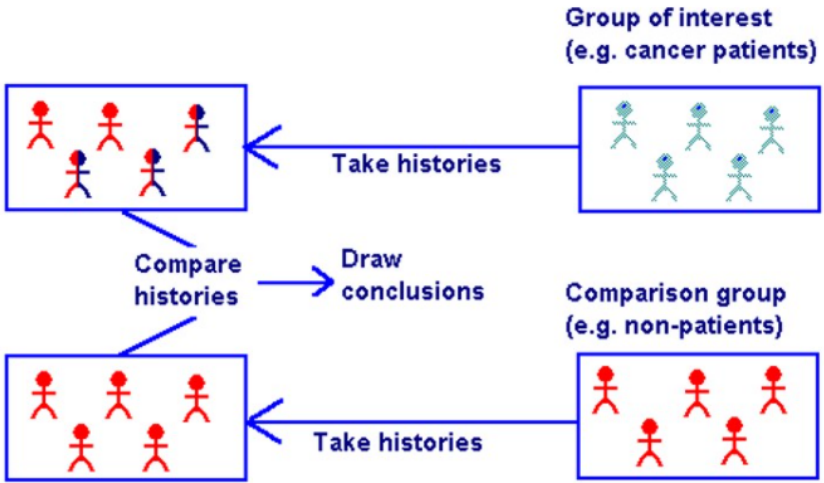
Advantage:

- Multiple outcomes can be studied
- Good for investigating rare exposures

Disadvantage:

- Time consuming

Case-Control Study



Case-control study

Advantage:

- Multiple exposures can be studied
- Good for examining rare outcomes or outcomes with long latency
- Relatively quick to conduct

Disadvantage:

- Recall bias can occur since exposure data is collected retrospectively
- Cannot identify risk in exposed/unexposed group

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Example: "Original" Job Training Program in 1976

- Related papers: LaLonde (1986); Dehejia and Wahba (1999)
- The National Supported Work Demonstration was a randomized experiment of a job training program conducted in the mid-1970s.
- The program was a temporary employment program designed to help disadvantaged workers lacking basic job skills move into the labor market by giving them work experience and counseling in a sheltered environment.
- Those assigned to the treatment group received all the benefits of the program, while those assigned to the control group were left to fend for themselves.
- We consider male participants in the program. 297 men were assigned to the treatment group and 425 to the control.

Check Covariate Balancing

- Two-sample t-test statistics:

	Treatment Mean	Control Mean	P-Value (Two-Sided)
age	24.626	24.447	0.722
education	10.380	10.188	0.144
black	0.801	0.800	0.965
hispanic	0.094	0.113	0.415
married	0.168	0.158	0.703
nodegree	0.731	0.814	0.009
earnings75	3066.098	3026.683	0.917

- Notice that, as expected, in a real randomization experiment the covariates are generally balanced
- For 6 of the 7 covariates examined, the difference between the two groups was not statistically significant at the 0.05 level. However, for the variable "no high school degree," the difference was statistically significant
- This result is consistent with what one might expect when conducting 7 independent significance tests purely by chance, i.e., with differences arising only from the random assignment of treatments

Job Training Program with Alterantive Controls

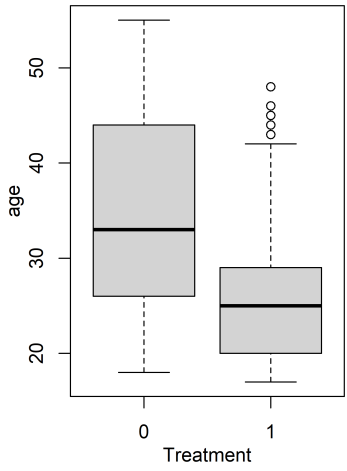
- LaLonde (1986); Dehejia and Wahba (1999): The treated group came from the original job training program experiment, but the control group was drawn from the Panel Study of Income Dynamics (PSID) dataset—not from the original experiment.
- The treatment is participation in the job training program. There are 185 men in the treated group with available earnings data in 1974, and 2,490 men in the control group. The job training occurred between 1976 and 1977, and the outcome of interest is earnings in 1978.

Check Covariate Balancing

- Two-sample t-test statistics:

	Treatment Mean	Control Mean	P-Value (Two-Sided)
age	25.82	34.76	0.0000
education	10.35	11.99	0.0000
black	0.84	0.26	0.0000
hispanic	0.06	0.03	0.1455
married	0.19	0.87	0.0000
nodegree	0.71	0.33	0.0000
earnings74	2095.57	20947.15	0.0000
earnings75	1532.06	20154.04	0.0000

- The covariates are imbalanced between the treated and control groups.
- Statistically, this implies $A \not\perp\!\!\!\perp X$, which is suggests that $A \not\perp\!\!\!\perp Y(a)$



Without Randomization: What Changes?

- Randomized experiments: Since $A \perp\!\!\!\perp Y(a)$:

$$\mathrm{E}\{Y(a)\} = \mathrm{E}\{Y(a)|A = a\} = \mathrm{E}(Y|A = a)$$

- Observational studies: Since $A \not\perp\!\!\!\perp Y(a)$:

$$\mathrm{E}\{Y(a)\} \neq \mathrm{E}\{Y(a)|A = a\} = \mathrm{E}(Y|A = a)$$

- This is confounding



- In order to identify $\mathrm{E}\{Y(a)\}$ in observational studies, we require a condition that plays a similar role to $A \perp\!\!\!\perp Y(a)$

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Causal Assumptions in Randomized Experiments

Assumption (SUTVA (Rubin, 1980))

- $Y = Y(A) = AY(1) + (1 - A)Y(0)$

- No interference between units (Cox, 1958)
- No different versions of a treatment; also known as (causal) consistency (Cole and Frangakis, 2009)

Assumption (Randomization)

- $\{Y(a), X\} \perp\!\!\!\perp A$, $\forall a$, *with* $\Pr(A = a) = p \in (0, 1)$

- A is independent of any pre-treatment variables

Causal Assumptions in Observational Studies

Assumption (SUTVA (Rubin, 1980))

- $Y = Y(A) = AY(1) + (1 - A)Y(0)$

Assumption (Conditional Ignorability)

- $Y(a) \perp\!\!\!\perp A \,|\, X, \forall a$

- Conditional on X, treatment assignment A is as if randomized
- aka Conditional exchangeability, No unmeasured confounding, No selection on unobservables, etc

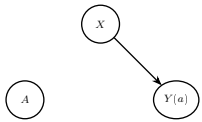
Assumption (Positivity/Overlap)

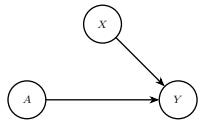
- $\Pr(A = a \,|\, X) \in (0, 1), \forall (a, X)$

- Every unit in the study has some probability of being assigned treament and control
- Rosenbaum and Rubin (1983): Strong ignorability = Conditional ignorability + Positivity

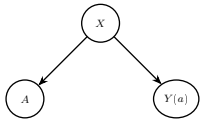
Relationship Between Randomization and Conditional Ignorability

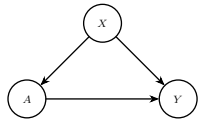
- Randomization : $\{Y(a), X\} \perp\!\!\!\perp A$





- Conditional ignorability : $Y(a) \perp\!\!\!\perp A \,|\, X$





  - X includes all confounders of A and Y; there are no unmeasured confounders

- Randomization $\Rightarrow$ Conditional ignorability

$$P\{Y(a)|A,X\} = \frac{P\{Y(a),A,X\}}{P(A,X)} = \frac{P\{Y(a),X\}P(A)}{P(X)P(A)} = \frac{P\{Y(a),X\}}{P(X)} = P\{Y(a)|X\}$$

- Conditional ignorability $\not\Rightarrow$ Randomization; many counterexamples

Connection to the Missing Data Framework

- For units with $A = 0$, the potential outcome $Y(1)$ is unobserved. Consequently, $Y(1)$ can be viewed as subject to a missing-data mechanism, with A serving as the missingness indicator.

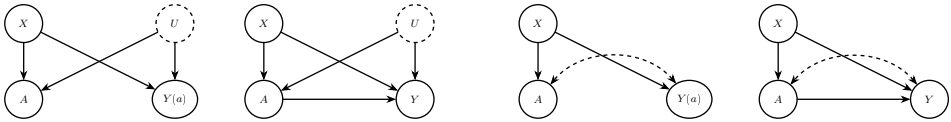
- **Missing completely at random (MCAR):** $Y(1) \perp\!\!\!\perp A$
  - Missing completely by chance; there is no systematic pattern in the missingness of $Y(1)$.

- **Missing at random (MAR):** $Y(1) \perp\!\!\!\perp A \mid X$
  - Example: Let $Y(1) =$ income, $X =$ wealth, and $A =$ indicator of reporting income.
  - MCAR would imply that reporting income does not depend on the income itself, which is often unrealistic.
  - Under MAR, individuals with lower wealth ($X = 0$) may be less likely to report their income ($A = 1$) than wealthier individuals ($X = 1$), creating a systematic pattern in missingness.
  - Within each subgroup defined by X, the missingness of $Y(1)$ can be considered random (i.e., MCAR within each level of X)

- Randomization = MCAR; Conditional ignorability = MAR

When Is Conditional Ignorability Vioated?

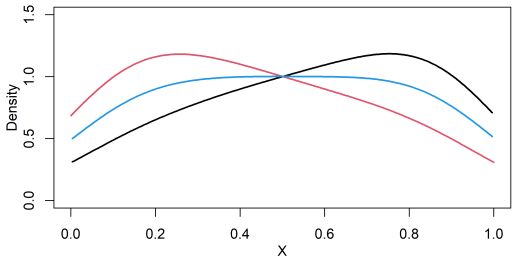
- Suppose there exists unmeasured confounder U
- Latent ignorability : $Y(a) \perp\!\!\!\perp A \,|\, X, U$



- Conditional ignorability $\nRightarrow$ Latent ignorability; Conditional ignorability $\nLeftarrow$ Latent ignorability
- **Missing Not at Random (MNAR):** $Y(1) \not\perp\!\!\!\perp A \,|\, X$
  - Even within subgroups defined by X, the missingness of $Y(1)$ cannot be considered random (i.e., not MCAR within each X)

Overlap Assumption and Transportability

- $\Pr(A = a \mid X) > 0 \quad \Leftrightarrow \quad \text{support}(X \mid A = 1) = \text{support}(X \mid A = 0)$
- This implies that the characteristics of treated units $(X \mid A = 1)$ and control units $(X \mid A = 0)$ "overlap"
- Consequently, even if $P(X \mid A = 1) \neq (X \mid A = 0)$, it is possible to transport information about one group to the other for any X, making the groups comparable (through appropriate weighting)
- Similarly, one can transport characteristics of both groups to the overall population
- Under randomization, $P(X \mid A = 1) = P(X \mid A = 0) = P(X)$



Black = $P(X|A = 1)$

Red = $P(X|A = 0)$

Blue = $P(X)$

Weak v. Strong Overlap

- Weak overlap: $\Pr(A = a \mid X) \in (0, 1)$ for all X
- Strong overlap: $\Pr(A = a \mid X) \in [\delta, 1 - \delta]$ for some $\delta > 0$
- Weak overlap is sufficient for identifying causal effects
- Strong overlap is required to ensure desirable asymptotic properties (e.g., $\sqrt{n}$-consistency and asymptotic normality) for estimators involving inverse propensity score weighting

Verifiability of Assumptions in Observed Data

- In general, it is impossible to verify whether the SUTVA and conditional ignorability assumptions hold
  - These assumptions concern potential outcomes $(Y(1), Y(0)$,
    yet the observed data (Y, A, X) provide no direct information about them
  - This is why attempting to estimate causal effects from data alone generally cannot yield valid causal conclusions, even when using fancy machine learning methods
  - To draw causal conclusions, SUTVA and conditional ignorability must be justified using subject-matter knowledge of the data-generating process.
    For example, if A is completely randomized within each X, we can be confident that these assumptions hold.
- However, the overlap assumption can be verified, since $\Pr(A|X)$ can be estimated from the observed data

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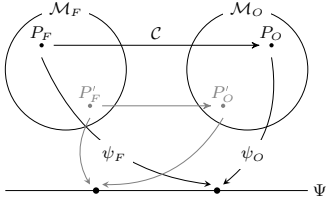
Identifiability

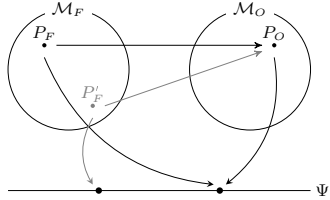
- $P_O(Y, A, X) \in \mathcal{M}_O$ be the observed data distribution; $P_F(Y(1), Y(0), A, X) \in \mathcal{M}_F$ be the full data distribution
  - P_O is a coarsened version of P_F; formally, there exists a mapping $\mathcal{C} : \mathcal{M}_F \to \mathcal{M}_O$ s.t. $P_O = \mathcal{C}(P_F)$
  - knowing P_F determines P_O, but the reverse is not necessarily true

- A parameter $\psi_F : \mathcal{M}_F \to \Psi$ is **identifiable** under a particular set of assumptions if

$$\exists \psi_O : \mathcal{M}_O \to \Psi \quad \text{s.t.} \quad \psi_F(P_F) = \psi_O(P_O) \ , \quad \forall P_F \in \mathcal{M}_F$$

We refer to ψ_O as the **identifying functional** of the parameter ψ_F





- Interpretation: a causal effect is
  - identifiable if the observed data distribution is compatible with a single causal effect value
  - nonidentifiable if the observed data distribution is compatible with multiple causal effect values

Example

- Under the SUTVA:

$$\begin{array}{ccc} P_F & -\mathcal{C}\rightarrow & P_O \\ X & = & X \\ A & = & A \\ Y(1), Y(0) & \rightarrow & Y = AY(1) + (1-A)Y(0) \end{array}$$

- Average treatment effect: $\psi_F : \mathcal{M}_F \rightarrow \mathbb{R}$

$$\psi_F(P_F) = \mathrm{E}\{Y(1) - Y(0)\} = \int y \, dP\{Y(1) = y\} - \int y \, dP\{Y(0) = y\}$$

- Average treatment effect on the treated: $\psi_F : \mathcal{M}_F \rightarrow \mathbb{R}$

$$\psi_F(P_F) = \mathrm{E}\{Y(1) - Y(0) | A = 1\} = \int y \, dP\{Y(1) = y | A = 1\} - \int y \, dP\{Y(0) = y | A = 1\}$$

- Conditional average treatment effect: $\psi_F : \mathcal{M}_F \rightarrow \{\text{fts of } X\}$

$$\psi_F(P_F) = \mathrm{E}\{Y(1) - Y(0) | X\} = \int y \, dP\{Y(1) = y | X\} - \int y \, dP\{Y(0) = y | X\}$$

What If These Conditions Fail?

- If these conditions fail, the observed data distribution is compatible with several values of the effect measure
- Example:

Unit	A	Y	$Y(1)$	$Y(0)$
1	0	1	1 zero	1
2	0	0	1 one	0
3	1	1	1	1 zero
4	1	0	0	1 one

Cond. Ign. Holds

$\tau_{\text{ATE}} = 0$

Unit	A	Y	$Y(1)$	$Y(0)$
1	0	1	0 zero	1
2	0	0	2 one	0
3	1	1	1	2 zero
4	1	0	0	0 one

Ign. Fails

$\tau_{\text{ATE}} = 0.5$

Unit	A	Y	$Y(1)$	$Y(0)$
1	0	1	2 zero	1
2	0	0	0 one	0
3	1	1	1	0 zero
4	1	0	0	2 one

Cond. Ign. Fails

$\tau_{\text{ATE}} = -0.5$

Robins' g-formula

Assumption

- *SUTVA:* $Y = Y(A) = AY(1) + (1 - A)Y(0)$
- *Conditional Ignorability:* $Y(a) \perp\!\!\!\perp A \,|\, X, \ \forall a$
- *Positivity/Overlap:* $\Pr(A = a \,|\, X) \in (0, 1), \ \forall (a, X)$

Robins' g-formula

Assumption

- *SUTVA:* $Y = Y(A) = AY(1) + (1 - A)Y(0)$
- *Conditional Ignorability:* $Y(a) \perp\!\!\!\perp A \,|\, X,\ \forall a$
- *Positivity/Overlap:* $\Pr(A = a \,|\, X) \in (0, 1),\ \forall (a, X)$

- Robins (1986): g-formula:

$$
\begin{aligned}
\mathrm{E}\{Y(a)\} &= \mathrm{E}_X\left[\mathrm{E}\{Y(a)|X\}\right] \\
&= \mathrm{E}_X\left[\mathrm{E}\{Y(a)|A = a, X\}\right] && \text{Cond. Ign. \& Positivity} \\
&= \mathrm{E}_X\{\mathrm{E}(Y|A = a, X)\} && \text{SUTVA} \\
&= \sum_x \mathrm{E}(Y|A = a, X = x)\Pr(X = x)
\end{aligned}
$$



COPSS Distinguished Achievement Award and Lectureship

James M. Robins Harvard University

For helping create the modern field of causal inference; for developing ground-breaking methods for causal inference; for the analysis of missing data; f r semi-and non-parametric models, and for the wide adoption of these methods in public health, clinical medicine, and the social sciences.

Google Scholar: James M. Robins (110K citations)

- Likewise, $\mathrm{E}\{Y(a)|X = x\} = \mathrm{E}(Y|A, X)$
- $\mathrm{E}(Y|A, X)$: outcome regression

Inverse Probability Weighted Formula

Assumption

- *SUTVA:* $Y = Y(A) = AY(1) + (1-A)Y(0)$
- *Conditional Ignorability:* $Y(a) \perp\!\!\!\perp A \,|\, X,\ \forall a$
- *Positivity/Overlap:* $\Pr(A = a \,|\, X) \in (0,1),\ \forall (a, X)$

- Horvitz and Thompson (1952); Rosenbaum and Rubin (1983): Inverse Probability Weighted (IPW) formula:

$$
\begin{aligned}
\mathrm{E}\{Y(a)\} &= \mathrm{E}_X\left[\mathrm{E}_{Y(a)}\{Y(a)\,|X\}\right] \\
&= \mathrm{E}_X\left[\mathrm{E}\left\{\frac{\mathbb{1}(A=a)}{\Pr(A=a|X)}\middle|X\right\}\mathrm{E}\{Y(a)|X\}\right] && \mathrm{E}\left\{\frac{\mathbb{1}(A=a)}{\Pr(A=a|X)}\middle|X\right\} = 1 \text{ \& Positivity} \\
&= \mathrm{E}_X\left[\mathrm{E}\left\{\frac{\mathbb{1}(A=a)Y(a)}{\Pr(A=a|X)}\middle|X\right\}\right] && \text{Cond. Ign.} \\
&= \mathrm{E}\left\{\frac{\mathbb{1}(A=a)Y}{\Pr(A=a|X)}\right\} && \text{SUTVA}
\end{aligned}
$$



Google Scholar: Paul R. Rosenbaum (96K citations)

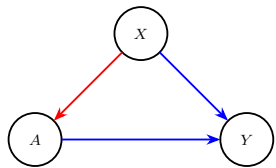
- $\Pr(A = a|X)$: propensity score

Identification of the ATE

- Under SUTVA, Cond. Ign, Positivity, we have

$$\begin{aligned}\tau_{\mathsf{ATE}} &= \mathrm{E}\{Y(1) - Y(0)\} \\ &= \mathrm{E}\left\{\frac{AY}{\Pr(A=1|X)} - \frac{(1-A)Y}{\Pr(A=0|X)}\right\} \qquad \text{IPW-based Representation} \\ &= \mathrm{E}\big\{\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X)\big\} \qquad \text{OR-based Representation}\end{aligned}$$

- The ATE can be identified either by knowing $\Pr(A|X)$ or $\mathrm{E}(Y|A,X)$
  - the treatment mechanism (i.e., how individuals receive the treatment)
  - the response mechanism (i.e., how individuals respond to the treatment)



Identification of the ATT: IPW

- We only need SUTVA, $Y(0) \perp\!\!\!\perp A|X$, $\Pr(A=1|X) < 1$:

$$\tau_{\mathsf{ATT}} = \mathrm{E}\{Y(1) - Y(0) \,|\, A=1\} = \frac{1}{\Pr(A=1)}\left[\mathrm{E}(AY) - \mathrm{E}\left\{\frac{\Pr(A=1|X)}{\Pr(A=0|X)}(1-A)Y\right\}\right]$$

- Note that

$$\begin{aligned}
\mathrm{E}\{Y(0)\,|A=1\} &= \frac{1}{\Pr(A=1)}\mathrm{E}\{AY(0)\} = \frac{1}{\Pr(A=1)}\mathrm{E}_X\left[\mathrm{E}\{AY(0)|X\}\right] \\
&= \frac{1}{\Pr(A=1)}\mathrm{E}_X\left[\mathrm{E}(A|X)\mathrm{E}\{Y(0)\,|X\}\right] \\
&= \frac{1}{\Pr(A=1)}\mathrm{E}_X\left[\Pr(A=1|X)\mathrm{E}\{Y(0)|X\}\right] \\
&= \frac{1}{\Pr(A=1)}\mathrm{E}_X\left[\frac{\Pr(A=1|X)}{\Pr(A=0|X)}\mathrm{E}(1-A|X)\mathrm{E}\{Y(0)|X\}\right] \\
&= \frac{1}{\Pr(A=1)}\mathrm{E}_X\left[\frac{\Pr(A=1|X)}{\Pr(A=0|X)}\mathrm{E}\{(1-A)Y(0)|X\}\right] \\
&= \frac{1}{\Pr(A=1)}\mathrm{E}_X\left[\frac{\Pr(A=1|X)}{\Pr(A=0|X)}\mathrm{E}\{(1-A)Y|X\}\right] = \frac{1}{\Pr(A=1)}\mathrm{E}\left\{\frac{\Pr(A=1|X)}{\Pr(A=0|X)}(1-A)Y\right\}
\end{aligned}$$

Identification of the ATT: OR

- We only need SUTVA, $Y(0) \perp\!\!\!\perp A|X$, $\Pr(A=1|X) < 1$:

$$\tau_{\text{ATT}} = \mathrm{E}\{Y(1) - Y(0) \,\big|\, A=1\} = \frac{1}{\Pr(A=1)}[\mathrm{E}(AY) - \mathrm{E}\{A\mathrm{E}(Y|A=0,X)\}]$$

- Note that

$$\begin{aligned}\mathrm{E}\{Y(0)|A=1\} = \frac{\mathrm{E}\{AY(0)\}}{\Pr(A=1)} &= \frac{\mathrm{E}_X\left[\mathrm{E}\{AY(0)|X\}\right]}{\Pr(A=1)} \\ &= \frac{\mathrm{E}_X\left[A\mathrm{E}\{Y(0)|A=1,X\}\right]}{\Pr(A=1)} = \frac{\mathrm{E}_X\left[A\mathrm{E}\{Y(0)|A=0,X\}\right]}{\Pr(A=1)} = \frac{\mathrm{E}_X\{A\mathrm{E}(Y|A=0,X)\}}{\Pr(A=1)}\end{aligned}$$

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Propensity Score

- The **propensity score** is defined as

$$e(X) = \Pr(A = 1 \mid X)$$

- First introduced by Rosenbaum and Rubin (1983): Biometrika's most cited paper ($>$ 40K citations)

- Given its central role in causal inference, we next discuss key properties of the propensity score

Balancing Score

- A balancing score $b(X) : \mathbb{R}^{\dim(X)} \to \mathbb{R}^q$ satisfies $X \perp\!\!\!\perp A \,|\, b(X)$
- The balancing score acts as a sufficient statistic: conditional on $b(X)$, X and A are independent
- Consequently, within each level of $b(X)$, the distribution of X is the same for treated and control groups, resembling a randomized experiment

Properties of the Propensity Score

- **Property 1**: The propensity score is the balancing score: $A \perp\!\!\!\perp X \,|\, e(X)$

- Useful fact: $\mathrm{E}\{V|W, f(W)\} = \mathrm{E}(V|X)$

- Note that

$$\Pr\{A = 1 \mid X, e(X)\} = \Pr(A = 1 \mid X) = e(X)$$

and

$$\Pr\{A = 1 \mid e(X)\} = \mathrm{E}\{A \mid e(X)\} = \mathrm{E}\left[\mathrm{E}\left\{A \mid e(X), X\right\} \mid e(X)\right] = \mathrm{E}\left\{\mathrm{E}\left(A \mid X\right) \mid e(X)\right\} = \mathrm{E}\left\{e(X) \mid e(X)\right\} = e(X)$$

- If two individuals have the same propensity score, then the probability that each receives the treatment is purely random

- We do not need the potentially high-dimensional covariates X to achieve balance;
  a one-dimensional $e(X)$ is sufficient

Properties of the Propensity Score

- **Property 2**: The propensity score is the optimal balancing score, i.e., for any balancing score $b(X)$, there exists a function f s.t. $e(X) = f\big(b(X)\big)$

- Suppose the claim is false, i.e., there is no function f such that $e(X) = f(b(X))$. This is equal to

$$\exists X \neq X' \text{ such that } b(X) = b(X') \text{ and } e(X) \neq e(X')$$

From the definition of the balancing score $A \perp\!\!\!\perp X | b(X)$, we get

$$\Pr\{A = 1|b(X)\} = \Pr\{A = 1|X, b(X)\} = \Pr(A = 1|X) = e(X)$$

and

$$\Pr\{A = 1|b(X')\} = \Pr\{A = 1|X', b(X')\} = \Pr(A = 1|X') = e(X')$$

But, we observe $\Pr\{A = 1|b(X)\} = \Pr\{A = 1|b(X')\}$ because $b(X) = b(X')$. This contradicts $e(X) \neq e(X')$.

- This property is conceptually similar to minimal sufficiency

Properties of the Propensity Score

- **Property 3**: $Y(a) \perp\!\!\!\perp A|X \Rightarrow Y(a) \perp\!\!\!\perp A|e(X)$

- Note that

$$\Pr\{A = 1 \mid X, e(X)\} = \Pr(A = 1 \mid X) = e(X)$$

and

$$\begin{aligned}
\Pr\{A = 1 \mid Y(a), e(X)\} &= \mathrm{E}\{A \mid Y(a), e(X)\} \\
&= \mathrm{E}\left[\mathrm{E}\left\{A \mid Y(a), e(X), X\right\} \mid Y(a), e(X)\right] \\
&= \mathrm{E}\left\{\mathrm{E}\left\{A \mid Y(a), X\right\} \mid e(X)\right\} \\
&= \mathrm{E}\left[\mathrm{E}\left(A \mid X\right) \mid e(X)\right] = \mathrm{E}\left\{e(X) \mid e(X)\right\} \\
&= e(X)
\end{aligned}$$

- Combined with Property 1, under cond. ign., we have $\{Y(a), X\} \perp\!\!\!\perp A|e(X)$

- Under cond. ign., A can be viewed as randomized within each level of $e(X)$ (one-dimensional); previously, we considered A randomized within each level of X (potentially high-dimensional)

Properties of the Propensity Score

- **Property 4**: $\tau_{\mathsf{ATE}} = \mathrm{E}[\mathrm{E}\{Y|A=1, e(X)\} - \mathrm{E}\{Y|A=0, e(X)\}]$

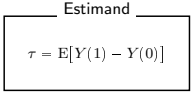
- From the positivity:

$$\begin{aligned}
&\Pr\{A=1, e(X)\} = \Pr\{A=1|e(X)\}P\{e(X)\} = e(X)P\{e(X)\} > 0\ , \\
&\Pr\{A=0, e(X)\} = \Pr\{A=0|e(X)\}P\{e(X)\} = \{1-e(X)\}P\{e(X)\} > 0
\end{aligned} \tag{1}$$

- Following the algebra in the g-formula:

$$\begin{aligned}
\mathrm{E}\{Y(a)\} &= \mathrm{E}_X[\mathrm{E}\{Y(a)|e(X)\}] \\
&= \mathrm{E}_X[\mathrm{E}\{Y(a)|A=a, e(X)\}] && \text{Property 2 \& (1)} \\
&= \mathrm{E}_X[\mathrm{E}\{Y|A=a, e(X)\}] && \text{SUTVA}
\end{aligned}$$

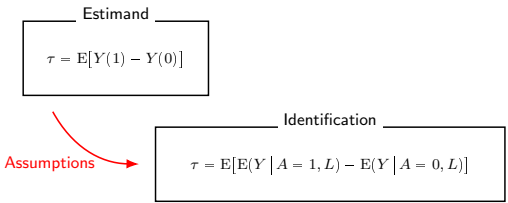
Flow of Causal Inference



Define an estimand

(e.g., Average Treatment Effect (ATE))

Flow of Causal Inference



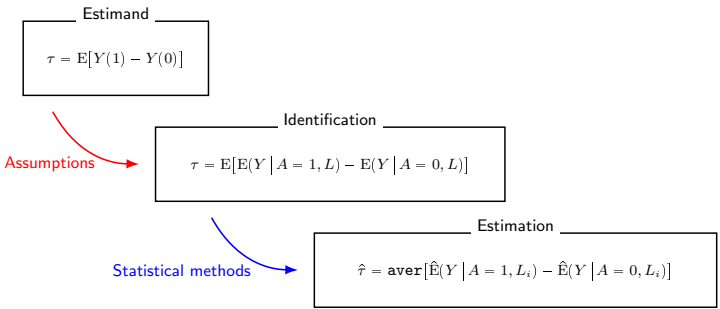
Define an estimand

(e.g., Average Treatment Effect (ATE))

Establish identification

(e.g., g-formula, IPW)

Flow of Causal Inference



Define an estimand

(e.g., Average Treatment Effect (ATE))

Establish identification

(e.g., g-formula, IPW)

Construct an estimator

(e.g., OLS, Machine learning)

References

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